Q04: RL-Based Reasoning vs. Ordinary Generation

How the optimization signal changes

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The question

Q04
How does RL-based reasoning differ from ordinary LLM generation?

- Generation: sample from a fixed policy.
- Training: change the policy using an objective.

Pre-training and inference

- Pre-training: next-token cross-entropy on a corpus.
- Inference: θ is fixed; prompt and decoding influence samples.
- Longer reasoning text can be elicited without a parameter update.

Post-training adds a target

SFT
imitate selected responses

Preference optimization
favor chosen over rejected responses

RL
maximize expected reward

These are mechanisms, not guarantees of reasoning.

Idealized RL objective

$$\max_{\theta} \ \mathbb{E}_{y\sim\pi_\theta}[r(x,y)] - \beta D_{\mathrm{KL}}(\pi_\theta \| \pi_{\mathrm{ref}})$$

- Reward says which outcome is preferred.
- KL term constrains policy drift.
- The practical result depends on the reward, data, and optimization.

Why verifiable rewards matter

- Tests, proof checkers, and numeric oracles can score outcomes.
- Process supervision can score intermediate steps.
- A verifier is only as good as its specification and coverage.

Reward what you can verify; audit what you failed to verify.

Do not equate reward with truth

- Reward hacking and shortcuts remain possible.
- A correct final answer need not imply a faithful rationale.
- Held-out, adversarial, and cost-controlled evaluation is required.

Toy experiment

- Sample one of several candidate answers.
- Give reward 1 only to the verifier-approved answer.
- Update probabilities; inspect how mass shifts.

A bandit demonstration, not RLHF or an LLM.

Takeaway

Core difference
Ordinary generation samples a learned distribution. RL-based post-training changes that distribution using a reward/preference objective. The reward defines the behavior being optimized—and its blind spots.

References: Ouyang et al. (2022); Rafailov et al. (2023); Lightman et al. (2023).